



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

27-02-2022

JEE(Main + Advanced) : Enthusiast Course (PHASE : TOAS, TNAS & TAAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	C	C	A	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,D	A,C	A,B,C,D	A,B,D	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.50	4.00	1.72 to 1.73	2.00	0.00	1.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	A	C	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	B,C,D	B,D	A,B,D	A,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	2.00	135.00	6.00	4.00	8.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	D	A	A	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	C,D	A,C,D	A,C	C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.50	4.00	6.00	1.50	2.00	2.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

$$f = +20$$

$$\therefore \frac{1}{v} - \frac{1}{-10} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{10} = \frac{-1}{20} = -20\text{cm}$$

$$\therefore m = \frac{-20}{-10} = 2$$

2. **Ans (C)**

$$E_1 = \frac{1}{2}kx^2$$

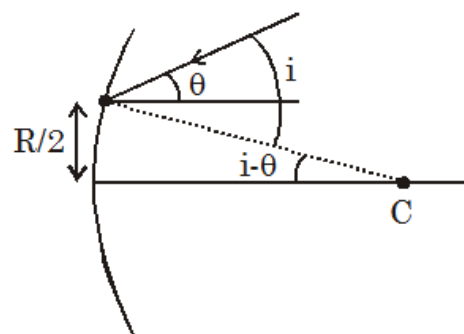
$$E_2 = \frac{1}{2}ky^2$$

$$E = \frac{1}{2}k(x+y)^2$$

3. **Ans (C)**

The ray will be totally reflected when angle of incidence is equal to critical angle.

$$\frac{\sin i}{\sin \frac{\pi}{2}} = \frac{\mu_2}{\mu_1} \Rightarrow \sin i = \frac{\mu_0}{\mu_1} = \frac{1}{2} \Rightarrow i = 30^\circ$$



From figure, $i - \theta = 30^\circ \Rightarrow \theta = 0$

4. **Ans (A)**

Both blocks loose contact immediately after the release.

$$T_P = 2\pi\sqrt{\frac{m}{4K}}, T_Q = 2\pi\sqrt{\frac{m}{K}}$$

$$\Rightarrow \therefore T_Q = 2T_P$$

Q comes at lowest position at time $\frac{T_Q}{2}$

travelling a distance $\frac{2mg}{K}$ downwards.

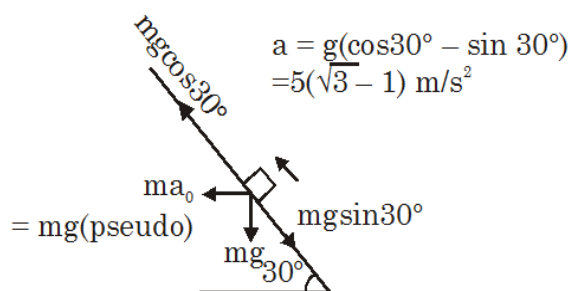
In time $\frac{T_Q}{2}$, i.e. time period of P (T_P) the block

P come back to original position

$\therefore$ The distance between P and Q is $\frac{2mg}{K}$

5. **Ans (B)**

w.r.t. incline (N.I.F.)



$$\therefore t_{CA} = \sqrt{\frac{2\ell}{a}} = \sqrt{\frac{2}{5(\sqrt{3} - 1)}}$$

$$= \sqrt{\frac{\sqrt{3} + 1}{5}} \approx \sqrt{\frac{2.7}{5}} \approx 0.74 \text{ sec}$$

6. **Ans (D)**

At $t = 3$

$$\text{Slope} = \frac{d\left(\frac{1}{V}\right)}{dt} = -1$$

$$\Rightarrow -\frac{1}{V^2} \frac{dV}{dt} = -1 \Rightarrow \left(\frac{1}{\sqrt{3}}\right)^2 \frac{dV}{dt} = 1$$

$$\Rightarrow a = \frac{dV}{dt} = 3 \text{ m/s}^2$$

PART-1 : PHYSICS

SECTION-I (ii)

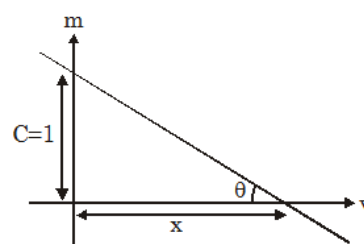
7. **Ans (A,B,C)**

$$m = \frac{f - v}{f}$$

$$m = 1 - \frac{v}{f}$$

$$\text{Slope} = -\frac{1}{f}$$

$$C = 1$$



$$\tan \theta = \frac{1}{x} = \frac{1}{f}$$

$$x = f$$

8. **Ans (A,D)**

$$K_{\max} = 16\pi^2 \quad T = 0.2 \text{ s}$$

$$f = \frac{1}{T} = 5 \text{ Hz}$$

$$\frac{1}{2}mv_{\max}^2 = K_{\max} \Rightarrow v_{\max} = 4\pi$$

$$A \frac{2\pi}{0.2} = 4\pi \Rightarrow A = 0.4 \text{ m}$$

At $t = 0.1$ sec the particle is at extreme position so acceleration is not zero.

9. **Ans (A,C)**

When velocity of object is perpendicular to mirror, relative velocity is maximum and

$$v_{\max} = 2 \text{ m/s.}$$

When velocity of object is parallel to mirror, relative velocity is minimum and $v_{\min} = 0 \text{ m/s}$

10. **Ans (A,B,C,D)**

$$\therefore F = \frac{-du}{dy} = 0 \quad (\text{For equilibrium})$$

Particle is performing simple harmonic motion

11. Ans (A,B,D)

(A, B) is moving with constant velocity then $a = 0$

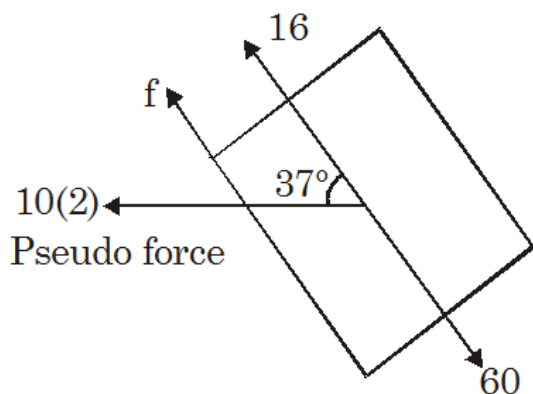
so friction available = $\mu mg \cos \theta$

$$= (0.8)(10)(10)(4/5) = 64 \text{ N.}$$

but $mg \sin \theta = 60 \text{ N}$

so required friction is 60 N .

So net force is zero.



$$(C) a = 2\hat{i} \quad f = mg \sin 37^\circ - ma \cos 37^\circ = 44 \text{ N}$$

$$(D) f = mg \sin 37^\circ - ma \cos 37^\circ = -20 \text{ N}$$

12. Ans (C,D)

$$\frac{dx}{dt} = 8x + 1$$

$$\int_0^x \frac{dx}{8x + 1} = \int_0^1 dt$$

$$\Rightarrow \ln(8x + 1) = 8t \Rightarrow x = \frac{1}{8}(e^{8t} - 1) \Rightarrow \frac{d^2x}{dt^2} > 0$$

at $t = 0$, $x = 0 \Rightarrow v = 1 \text{ m/s}$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.50)

Information based.

2. Ans (4.00)

$$T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{2}{5} \frac{mR^2}{C}}$$

$$I = \frac{2}{5} mR^2$$

3. Ans (1.72 to 1.73)

$$2(60 - r) = [2(60 - r) + 180 - 2r] \frac{1}{3}$$

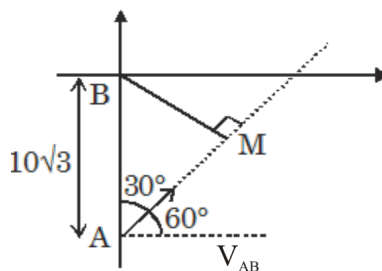
$$6(60 - r) - (60 - r) = 180 - 2r$$

$$4(60 - r) = 180 - 2r$$

$$60 = 2r \Rightarrow r = 30^\circ$$

$$\mu = \sqrt{3}$$

6. Ans (1.50)



As observed by B motion of A is along AM and

BM is the shortest distance between them.

Relative displacement of A w.r.t to B is $AM = AB$

$$\cos 30^\circ \vec{V}_{AB} = 5\hat{i} + 5\sqrt{3}\hat{j}$$

Time taken

$$(t) = \frac{|\vec{S}_{AB}|}{|\vec{V}_{AB}|} = \frac{AM}{V_{AB}} = \frac{10\sqrt{3} \cos 30^\circ}{10} = 1.5 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (A)

More no. of particle, implies more depression in freezing point hence lesser freezing point.

4. Ans (C)

$$\text{Net Na}^+ = 4 - \left(2 \times \frac{1}{4} + 1\right) = 5/2, \text{Net Cl}^- = 4,$$

$$\text{PF} = \frac{\frac{5}{2} \times \frac{4}{3} \pi r_+^3 + 4 \times \frac{4}{3} \pi r_-^3}{16\sqrt{2} r_-^3}$$

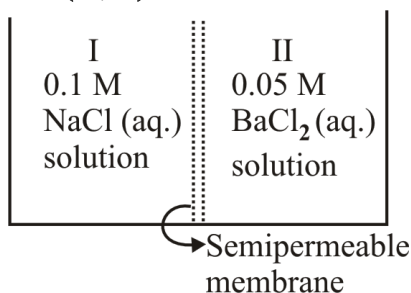
$$\sqrt{2} a = 4r_-$$

$$a = \frac{4r_-}{\sqrt{2}} = 2\sqrt{2} r_-$$

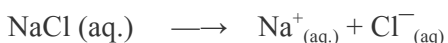
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (B,D)



In partition I



$$C = 0.1\text{M} \quad 0.1\text{M} \quad 0.1\text{M}$$

$$i = 1 + (n-1) \propto \quad (n = \text{no. of ions})$$

$$i = 1 + (2-1) \times 1 \quad \text{assuming 100\% dissociation}$$

$i = 2$ $\text{NaCl \& BaCl}_2 \rightarrow$ strong electrolyte

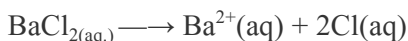
$$\text{So,} \quad \% \alpha = 100\%$$

$$\pi_I = iCRT \quad \boxed{\alpha = 1}$$

$$\pi_I = 2 \times 0.1 \times R \times T$$

$$\pi_I = 0.2 RT$$

In partition II



$$C = 0.05\text{M} \quad 0.05\text{M} \quad 2 \times 0.05\text{M}$$

$$n = 3 \text{ and } \alpha = 1$$

$$i = 1 + (3-1) \times 1 \Rightarrow \boxed{i = 3}$$

$$\begin{aligned} \text{So, } \pi_{II} &= i \times CRT \\ &= 3 \times 0.05 \times RT \\ &= 0.15 RT \end{aligned}$$

$$\begin{aligned} \text{So, } \pi_I &= 0.20 RT \\ \text{and } \pi_{II} &= 0.15 RT \end{aligned}$$

$$\text{hence } \pi_I > \pi_{II}$$

So, Solution in partition I is hypertonic. Due to that movement of water from BaCl_2 will flow towards the NaCl solution.

(B) option (B) is correct

$$\pi_I (\text{NaCl}) > \pi_{II} (\text{BaCl}_2)$$

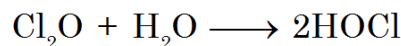
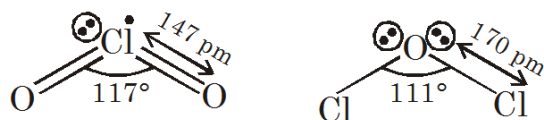
(D) option (D) is correct

$$\text{So, } \boxed{\text{Ans : - (B) \& (D)}}$$

10. Ans (A,B,D)

Part of theory

11. Ans (A,C,D)



12. Ans (A,B,D)

The order of solubility of

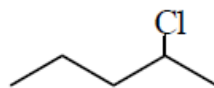
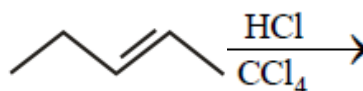
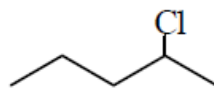
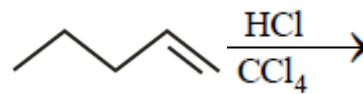
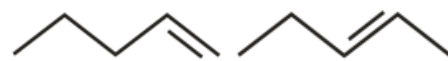


can only be explained on the basis of lattice energy rest others can be explained by Fajan's rule.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (2.00)

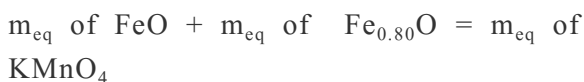


3. Ans (135.00)

let mmoles of each is = x

n-factor of $\text{FeO} = 1$

n-factor of $\text{Fe}_{0.80}\text{O} = 0.4$



$$x \times 1 + x \times 0.4 = 70 \times 0.3 \times 5$$

$$x = 75 \text{ mmoles}$$

$$\begin{aligned} \text{mmoles of Fe}^{3+} \text{ produced} &= 75 + 75 \times 0.8 \\ &= 135 \text{ mmoles} \end{aligned}$$

5. **Ans (4.00)**

Refer to notes.

6. **Ans (8.00)**

Refer to notes.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

LHS = +ve & RHS = -ve therefore solution is not possible.

2. **Ans (D)**

$\cot^{-1}x > 0$ for all x

$$|\tan^{-1}x| + \cot^{-1}x = \frac{\pi}{2}$$

$$|\tan^{-1}x| = \left(\frac{\pi}{2} - \cot^{-1}x\right)$$

$$|\tan^{-1}x| = \tan^{-1}x$$

valid when $\tan^{-1}x \geq 0$ is possible when $x \geq 0$

3. **Ans (A)**

Multiplying both sides by $(e^x - 1)$

$$f(2x)(e^x - 1) = (e^{2x} - 1)f(x)$$

$$\frac{f(2x)}{e^{2x} - 1} = \frac{f(x)}{e^x - 1} = \frac{f(x/2)}{e^{x/2} - 1} = \dots = \frac{f(x/2^n)}{e^{x/2^n} - 1}$$

$$\frac{f(x)}{e^x - 1} = \lim_{n \rightarrow \infty} \frac{f(x/2^n)}{e^{x/2^n} - 1} = \lim_{h \rightarrow 0} \frac{f(h)}{h} \times \frac{h}{e^h - 1}$$

$$\frac{f(x)}{e^x - 1} = f'(0)$$

$$f(x) = e^x - 1$$

$$R_f : (-1, \infty)$$

4. **Ans (A)**

$$F(x) = (\sin x - \cos x)^2 + 2(\sin x - \cos x) + 3$$

$$F_{\max}(x) = (\sqrt{2})^2 + 2\sqrt{2} + 3 = 5 + 2\sqrt{2}$$

5. **Ans (B)**

$$S = \frac{1}{1^5} + \frac{1}{3^5} + \frac{1}{5^5} + \frac{1}{7^5} + \dots$$

$$= \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \frac{1}{4^5} + \dots\right) - \left(\frac{1}{2^5} + \frac{1}{4^5} + \frac{1}{6^5} + \dots\right)$$

$$= (k) - \frac{1}{32} \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \dots\right)$$

$$= k - \frac{k}{32} = \frac{31k}{32}$$

6. **Ans (D)**

$$S_n = 3n^3 + 2n^2$$

$$T_6 = S_6 - S_5 = (720) - (425) = 295$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A,C,D)**

$$T_n = (2^n) + (n^2 + 1)$$

$$S_n = \sum_{n=1}^n T_n$$

$$= \sum_{n=1}^n (2^n) + \sum_{n=1}^n (n^2) + \sum_{n=1}^n (1)$$

$$= \frac{2(2^n - 1)}{2 - 1} + \frac{n(n+1)(2n+1)}{6} + n$$

$$= 2^{n+1} + \frac{n^3}{3} + \frac{n^2}{2} + \frac{7n}{6} - 2$$

$$A = 1, B = \frac{1}{3}, C = \frac{1}{2}, D = \frac{7}{6}, E = -2$$

8. **Ans (C,D)**

$$x^2 + \frac{27}{x^2} \geq 2\sqrt{27}$$

$$x^2 + \frac{27}{x^2} + 1 \geq 6\sqrt{3} + 1$$

$$6\sqrt{3} + 1 \approx 11.4$$

9. **Ans (A,C,D)**

$$\sqrt{(|x|-2)(|x|-3)} \text{ is real if } 0 \leq |x| \leq 2 \text{ or } |x| \geq 3$$

$$\sqrt{8+2|x|-|x|^2} = \sqrt{-(|x|-4)(|x|+2)}$$

$$\text{is real if } 0 \leq |x| \leq 4$$

$$\Rightarrow f(x) \text{ is real for } 0 \leq |x| \leq 2 \text{ or } 3 \leq |x| \leq 4$$

$$\Rightarrow x \in [-2, 2] \cup [3, 4] \cup [-4, -3]$$

10. **Ans (A,C)**

$$\sin^{-1} \left(x^2 - 4x + \frac{7}{2} \right) = \frac{\pi}{6}$$

$$x^2 - 4x + \frac{7}{2} = \frac{1}{2}$$

$$x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

11. **Ans (C,D)**

Let

$$\tan^{-1} \frac{x}{2} = \theta \Rightarrow \tan \theta = \frac{x}{2}; \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cos\left(\tan^{-1} \frac{x}{2}\right) = \cos \theta = \frac{2}{\sqrt{4+x^2}}$$

$$\tan\left(\sin^{-1}\left(\frac{2}{\sqrt{4+x^2}}\right)\right) = \tan \tan^{-1}\left(\frac{2}{|x|}\right) = \frac{2}{|x|}$$

 12. **Ans (A,B,C,D)**

If base and number both are greater than one, then value of log is always positive.

PART-3 : MATHEMATICS

SECTION-II

 1. **Ans (2.50)**

Put $x = 3$ and $x = 7$ in given equation

to get $f(3) = 11$ & $f(7) = -1$

$$2f(3) + 4f(7) = 18$$

$$f(7) + 2f(3) = 21$$

$$\left[\frac{f(3)}{2f(7)}\right] = 5$$

 2. **Ans (4.00)**

$$x^2 + kx + k \geq 0 \quad \forall x \in \mathbb{R}$$

$$D \leq 0 \Rightarrow k \in [0, 4] \quad \dots(1)$$

$$\text{Also, } x^2 + k \neq 0 \quad \forall x \in \mathbb{R}$$

$$(1) \cap (2)$$

$$k \in (0, 4]$$

 3. **Ans (6.00)**

$$\tan^{-1}\left(\frac{\frac{1}{x} + \frac{1}{y}}{1 - \frac{1}{xy}}\right) = \tan^{-1}\left(\frac{1}{7}\right) \Rightarrow x = 7 + \frac{50}{y-7}$$

positive integral values of y for which $x \in \mathbb{I}^+$

are 8, 9, 12, 17, 32, 57.

 4. **Ans (1.50)**

$$\text{For } x \geq \frac{3}{2}; \theta = \pi$$

$$\therefore \frac{\cos \pi + \sin \pi + 4}{\sec \pi} = \frac{-1 + 4}{-1} = -3$$

 5. **Ans (2.00)**

$$\sec(2\theta - 2\alpha), \frac{1}{2} \operatorname{cosec}^2 \theta, \sec(2\theta + 2\alpha) \text{ are in HP}$$

$$\Rightarrow \cos(2\theta - 2\alpha) + \cos(2\theta + 2\alpha) = 4\sin^2 \theta$$

$$\cos 2\theta(1 + \cos 2\alpha) = 1$$

$$\Rightarrow 2\cos^2 \alpha = \sec 2\theta$$

$$\sec 2\theta \sec^2 \alpha = 2$$

 6. **Ans (2.00)**

$$b^2 + 4b + 4 = 1 + (a + c) + ac$$

$$b^2 + 2b + 3 = ac$$

$$ac = (b + 1)^2 + 2$$

$$\min. = 2$$



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JEE (Advanced)

REVIEW TEST

27-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS & TAAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,C,D	B,C,D	A,B,C	A,B	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.40	15.00	0.37 to 0.40	25.00	2.50	7.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	9	3	7	3	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C,D	A,D	A,B,C	B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	6.00	374.15	2.44 to 2.45	3.00	5.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	4	4	2	1	4

PART-3 : MATHEMATICS

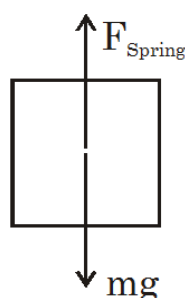
SECTION-I	Q.	1	2	3	4	5	6
	A.	C,D	A,B	A	A,B,C	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.50	0.00	7.00	2.50	6.00	7.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	1	2	2	7	1

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A,B,D)



Speed will be maximum when the block passes through the equilibrium position.

$$F_{\text{spring}} = mg$$

∴ A is correct

(B) From conservation of energy

$$mg(\ell + x) = \frac{1}{2}Kx^2 + \frac{1}{2}mv_{\text{max}}^2 \quad \dots (i)$$

$$\text{Where } x = \frac{mg}{K} = \frac{mg}{4mg}\ell = \frac{\ell}{4}$$

putting $x = \frac{\ell}{4}$ in equation (i)

$$V_{\text{max}} = \frac{3}{2}\sqrt{g\ell}$$

(C) is incorrect as in equilibrium

compression $\frac{\ell}{4}$ which is more than $\frac{\ell}{8}$

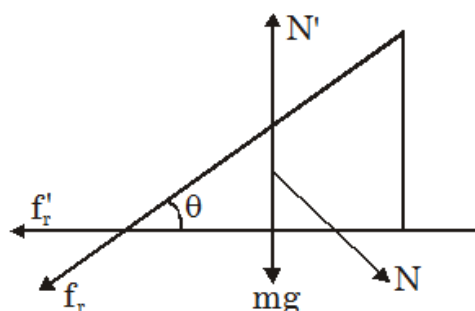
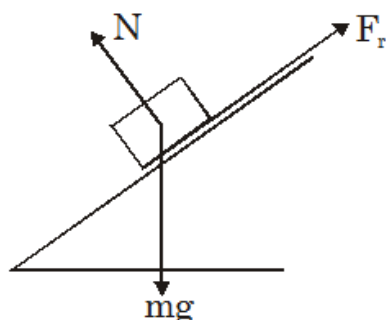
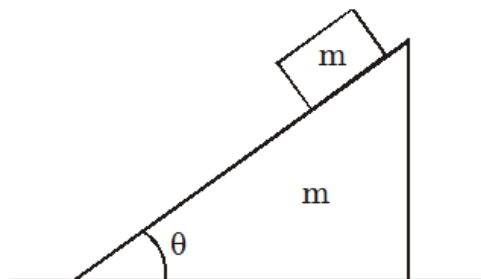
$$(D) \text{ Time taken is } \frac{T}{4} = \frac{1}{4}2\pi\sqrt{\frac{m}{K}}$$

∴ D is correct

2. **Ans (A,B,C,D)**

The above lens mirror combination is an equivalent converging mirror.

3. **Ans (B,C,D)**

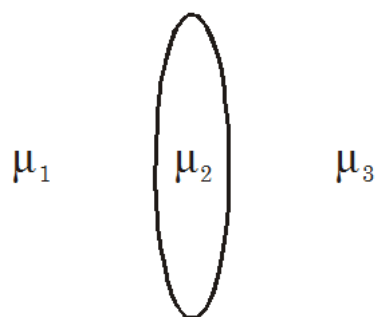


$$N' = mg + N \cos \theta + f_r \sin \theta$$

$$N \sin \theta = f_r + f_r \cos \theta$$

(for inclined plane to move over horizontal surface)

4. **Ans (A,B,C)**

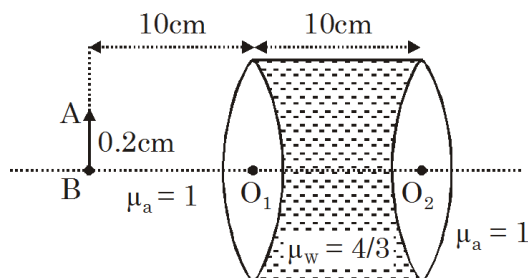


$$\frac{\mu_3}{V} - \frac{\mu_1}{U} = \frac{\mu_2 - \mu_1}{R_2} + \frac{\mu_2 - \mu_1}{R_1}$$

5. **Ans (A,B)**

As the prism are identical so deviation will be zero for even number of prism and δ for odd number of prism.

6. **Ans (A,C)**



For the image formed by the first lens

$$\begin{aligned} \frac{\mu_3}{v} - \frac{\mu_1}{u} &= \left(\frac{\mu_2 - \mu_1}{R_1} \right) + \left(\frac{\mu_3 - \mu_2}{R_2} \right) \\ \Rightarrow \frac{4}{3v} - \frac{1}{(-10)} &= \frac{\left(\frac{3}{2} - 1 \right)}{10} + \frac{\left(\frac{4}{3} - \frac{3}{2} \right)}{-10} \\ \frac{4}{3v} + \frac{1}{10} &= \frac{1}{20} + \frac{1}{60} \\ \Rightarrow \frac{4}{3v} &= \frac{1}{15} - \frac{1}{10} \end{aligned}$$

$$\Rightarrow v = -40 \text{ cm}$$

Lateral magnification,

$$m_1 = \frac{\mu_1}{\mu_3} \frac{v}{u} = \frac{1 \times 3}{4} \left(\frac{-40}{-10} \right) = +3$$

For the image formed by the second lens

$$\begin{aligned} \frac{\mu_3}{v} - \frac{\mu_1}{u} &= \left(\frac{\mu_2 - \mu_1}{R_1} \right) + \left(\frac{\mu_3 - \mu_2}{R_2} \right) \\ \Rightarrow \frac{1}{v} - \frac{4}{3(-50)} &= \left(\frac{\frac{3}{2} - \frac{4}{3}}{10} \right) + \left(\frac{1 - \frac{3}{2}}{-10} \right) \\ \Rightarrow \frac{1}{v} + \frac{2}{75} &= \frac{1}{60} + \frac{1}{20} \\ \Rightarrow \frac{1}{v} &= \frac{1}{15} - \frac{2}{75} \end{aligned}$$

$$\Rightarrow v = +25 \text{ cm}$$

The position of final image formed is 25 cm right to the second lens.

Lateral magnification,

$$m_2 = \frac{\mu_1}{\mu_3} \frac{v}{u} = \frac{4}{3 \times 1} \left(\frac{25}{-50} \right) = -\frac{2}{3}$$

Overall magnification,

$$m = m_1 \times m_2 = (+3) \times \left(-\frac{2}{3} \right) = -2$$

The size of the final image formed = $2 \times 0.2 \text{ cm}$
= 0.4 cm

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.40)**

$$V_x = 180 \cos 60^\circ = 9 \text{ m/s.}$$

$$V_{ly} = \sqrt{15^2 - 9^2} = 12 \text{ m/s}$$

$$V_{2y} = -12 \text{ m/s}$$

$$\therefore \Delta t = \frac{-12 - (12)}{-10} = \frac{12}{5} \text{ sec.}$$

2. **Ans (15.00)**

$$V_B \cos \theta_B = V_A \cos \theta_A \Rightarrow V_B = 15 \text{ m/s}$$

3. **Ans (0.37 to 0.40)**

Applying Snell's law between the points O and

P, we have

$$2 \times \sin 60^\circ = (\sin 90^\circ) \times \frac{2}{(1 + H^2)},$$

$$2 \times \frac{\sqrt{3}}{2} = 1 \times \frac{2}{(1 + H^2)}$$

$$(1 + H^2) = \frac{2}{\sqrt{3}}, H = \sqrt{\left(\frac{2}{\sqrt{3}} - 1\right)}$$

4. **Ans (25.00)**

$$\overrightarrow{PQ} = \sqrt{15^2 + 20^2} = 25 \text{ m}$$

$$25 = \frac{V_P^2 \sin 2(45)}{g} \Rightarrow V_P = 5\sqrt{10} \text{ m/s}$$

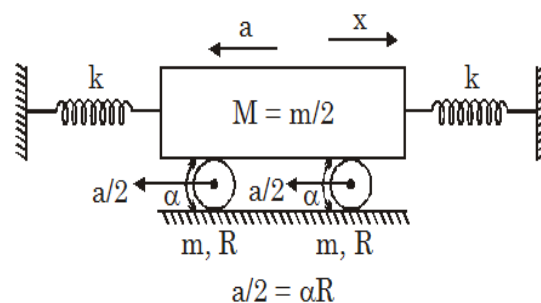
$$\therefore V_A^2 \sin^2 \theta - 2 \times 10 \times 12.5 = (5\sqrt{5})^2$$

$$\Rightarrow V_A \sin \theta = 5\sqrt{15} \text{ m/s}$$

$$V_A \cos \theta = V_P \cos 45^\circ = 5\sqrt{5} \text{ m/s and } \theta = 60^\circ$$

$$\therefore \overrightarrow{AB} = \frac{V_A^2 \sin 2\theta}{g} = 25\sqrt{3}$$

5. **Ans (2.50)**



$$2kx - 2f_1 = Ma = \frac{ma}{2} \dots\dots(i)$$

$$f_1 + f_2 = \frac{ma}{2} \dots\dots(ii)$$

$$(f_1 - f_2) R = \frac{mR^2}{2} \alpha,$$

$$(f_1 - f_2) = \frac{m\alpha R}{2} = \frac{ma}{4} \dots\dots(iii)$$

Adding (i), (ii) and (iii), we get

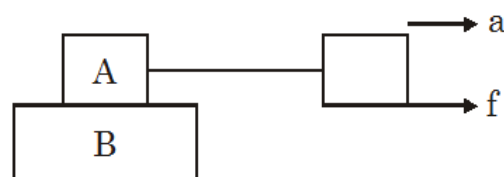
$$2kx = \frac{5ma}{4} \Rightarrow a = \frac{8kx}{5m}$$

$$\frac{d^2x}{dt^2} = - \left(\frac{8k}{5m} \right) x \Rightarrow T = 2\pi \sqrt{\frac{5m}{8k}} = \pi \sqrt{\frac{5m}{2k}}$$

6. **Ans (7.50)**

If all blocks moves with same acceleration

$$a = \frac{F}{2M + 2m}$$



$$f = (m + m + M) a = \frac{(2m + M) F}{2(m + M)}$$

For no slipping $f \leq f_1$

$$\frac{(2m + M)}{2(m + M)} F \leq \mu_s mg$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (6)**

$$\frac{1}{f_L} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{20} + \frac{1}{-20}\right)$$

$$\Rightarrow f_L = 20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-10} - \frac{2}{20} = -\frac{1}{5}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-5} \Rightarrow \frac{1}{30} - \frac{1}{5}$$

$$\text{For lens } \frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

2. **Ans (9)**

$$a = \mu g$$

$$6^2 = 0^2 + 2 \times \mu g S$$

3. **Ans (3)**

$$\therefore \mu = \frac{\sin\left(A + \frac{\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{A + \frac{A}{2}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{3A}{4}\right)}{\sin\left(\frac{A}{2}\right)}, \text{ Let } \frac{A}{4} = \theta$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin \theta - 4\sin^3 \theta}{2 \sin \theta \cdot \cos \theta} = \frac{3 - 4\sin^2 \theta}{2 \cos \theta}$$

$$\Rightarrow 4\sqrt{3}\cos^2 \theta - 4 \cos \theta - \sqrt{3} = 0$$

$$\Rightarrow \cos \theta = \frac{1 \pm 2}{2\sqrt{3}} = \frac{\sqrt{3}}{2}, \left(-\frac{1}{2\sqrt{3}}\right)$$

$$\Rightarrow \theta = 30^\circ \Rightarrow \frac{A}{4} = 30^\circ = \frac{\pi}{6}$$

$$\Rightarrow A = \frac{2\pi}{3}$$

$$\therefore C = 3$$

PART-2 : CHEMISTRY

SECTION-II

3. **Ans (374.15)**

$$\Delta T_b = K_b \cdot m = 0.52 \times \frac{11.7}{58.5} \times 2 + \frac{34.2}{342} = 1K$$

$$\therefore \text{B.P.} = 373.15 + 1 = 374.15 \text{ K.}$$

PART-2 : CHEMISTRY

SECTION-III

3. **Ans (4)**

$$\frac{r_+}{r_-} = \frac{1.6}{1.864} = 0.858$$

So, it is CsCl type unit cell

$$\text{So } \sqrt{3} a = 2(r_+ + r_-)$$

$$\text{So } a = \frac{2(1.864 + 1.6)}{\sqrt{3}} \text{ \AA} = 2 \times 2 \text{ \AA} = 4 \text{ \AA}$$

4. **Ans (2)**

2 ppm

$$M_1 V_1 = M_2 V_2$$

$$50 \times M_1 = \frac{1}{50} \times 0.1$$

$$M_1(\text{HCO}_3^-) = \frac{1}{25 \times 10^3}$$

$$\text{m moles of } \text{HCO}_3^- = \frac{1}{25 \times 10^3} \times 50 = \frac{1}{500}$$

$$\text{m moles of } \text{Ca}^{2+} = \frac{1}{1000}$$

$$\text{m moles of } \text{CaCO}_3 = \frac{1}{1000}$$

$$\text{weight of } \text{CaCO}_3 = \frac{1}{1000} \times 10^{-3} \times 100 = 10^{-4}$$

$$\text{hardness} = [\text{weight of } \text{CaCO}_3 / \text{weight of water} \times 10^6]$$

$$= \frac{10^{-4}}{50} \times 10^6 = 2 \text{ ppm}$$

5. **Ans (1)**



6. **Ans (4)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C,D)**

$$f(f(x)) = \frac{2f(x) + 1}{f(x)} = \frac{\frac{2(2x+1)}{x} + 1}{\frac{2x+1}{x}} = \frac{5x+2}{2x+1}$$

$$g(x) = f(f(x)) - \frac{1}{f(x)}, x \neq 0, -\frac{1}{2}$$

$$g(x) = \frac{5x+2}{2x+1} - \frac{x}{2x+1}$$

$$g(x) = 2$$

$\Rightarrow g(x)$ is a constant function

2. **Ans (A,B)**

$$\sin^{-1}(x^2 - 5x + 7) = \sin^{-1}1$$

$$x^2 - 5x + 7 = 1$$

$$x^2 - 5x + 6 = 0$$

$$x = 2, 3$$

options (A) & (B) are correct.

3. **Ans (A)**

$$4\{x\} = x + [x]$$

The above equation has 2 solutions i.e. $x = 0$ & $x = 5/3$

4. **Ans (A,B,C)**

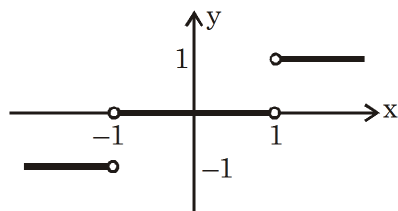
$$\text{Here, } b = \frac{2ac}{a+c} \quad \dots(1)$$

$$\text{and } b^2 = -2ac \quad \dots(2)$$

$$\Rightarrow b = \frac{-b^2}{a+c} \Rightarrow a+b+c=0$$

Now, verify it

5. **Ans (A,B,C,D)**



See answer from graph

6. **Ans (A,B,C)**

$$T_n = 100 + (n-1)(-5) = 0$$

$$\Rightarrow n = 21$$

$$\Rightarrow T_{21} = 0$$

$$\Rightarrow T_{21-5} = -T_{21+5} \Rightarrow T_{16} + T_{26} = 0$$

$$\text{Similarly } T_{11} + T_{31} = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.50)**

$$f^{-1}(x) = \frac{x-2}{2x-1} \Rightarrow f(x) = f^{-1}(x)$$

$$\Rightarrow f(f(x)) = x$$

$$\Rightarrow f(x) + 2f^{-1}(x) + 2 = f(f(x)) \text{ reduces to}$$

$$\frac{3(x-2)}{2x-1} = x-2$$

$$\Rightarrow \frac{(x-2)}{(2x-1)}(3-2x+1) = 0$$

$$\Rightarrow 2(x-2)(2-x) = 0 \Rightarrow x = 2$$

2. **Ans (0.00)**

$$\sin^{-1}x > 0 \text{ \& } \cos^{-1}x > 0$$

$$0 < x \leq 1 \text{ \& } -1 \leq x < 1$$

$$\Rightarrow \text{domain } (0,1)$$

Hence '0' integral values.

3. **Ans (7.00)**

$$f(x) = (\tan^{-1}x + \cot^{-1}x)^2 - 2\tan^{-1}x \cot^{-1}x$$

$$= \left(\frac{\pi}{2}\right)^2 - 2(\tan^{-1}x) \left(\frac{\pi}{2} - \tan^{-1}x\right)$$

$$= 2(\tan^{-1}x)^2 - \pi \tan^{-1}x + \frac{\pi^2}{4}$$

$$= 2\left(\tan^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8}$$

$$\therefore \text{Minimum value } \frac{\pi^2}{8} \text{ at } \tan^{-1}x = \frac{\pi}{4}$$

4. **Ans (2.50)**

$$\text{Put } \frac{2x}{3} = \tan \theta$$

then

$$2\sin^{-1}(\cos 2\theta) + 3\cos^{-1}(\sin 2\theta) + \tan^{-1} \tan 2\theta = \lambda \frac{\pi}{2}$$

$$\Rightarrow 2\left(\frac{\pi}{2} - \cos^{-1} \cos 2\theta\right) + 3\left(\frac{\pi}{2} - \sin^{-1} \sin 2\theta\right)$$

$$+ \tan^{-1} \tan 2\theta = \frac{\lambda\pi}{2}$$

$$\text{for } x = \frac{-17}{13} \Rightarrow \frac{-\pi}{2} < 2\theta < 0$$

$$\therefore 2\left(\frac{\pi}{2} + 2\theta\right) + 3\left(\frac{\pi}{2} - 2\theta\right) + 2\theta = \frac{\lambda\pi}{2}$$

$$\frac{5\pi}{2} = \frac{\lambda\pi}{2} \Rightarrow \lambda = 5$$

5. **Ans (6.00)**

$$\text{Given } a(1+r+r^2) = 21 \quad \dots(1)$$

$$\& a^2(1+r^2+r^4) = 189 \quad \dots(2)$$

$$\frac{\text{Eq. (2)}}{\text{Eq. (1)}^2} \Rightarrow \frac{r^2 - r + 1}{r^2 + r + 1} = \frac{3}{7}$$

$$\Rightarrow 7r^2 - 7r + 7 = 3r^2 + 3r + 3$$

$$\Rightarrow 4r^2 - 10r + 4 = 0$$

$$\Rightarrow 2r^2 - 5r + 2 = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r-2) - 1(r-2) = 0$$

$$\Rightarrow (\therefore r \notin \mathbb{I})$$

$$\therefore (1) \Rightarrow a\left(1 + \frac{1}{2} + \frac{1}{4}\right) = 21 \Rightarrow a = 12$$

$$\therefore S = \frac{a}{1-r} = \frac{12}{1-\frac{1}{2}} = 24 \Rightarrow \frac{S}{4} = 6$$

6. Ans (7.00)

$$2 \sin^2 x + 3 \operatorname{cosec}^2 x + a^2 + \frac{1}{a^2} \geq 7$$

using (A.M > G.M)

PART-3 : MATHEMATICS

SECTION-III

1. Ans (8)

$$f(x) + f(x+2) = f(x+1) \quad \dots\dots(i)$$

$$\Rightarrow f(x+1) + f(x+3) = f(x+2) \quad \dots\dots(ii)$$

adding (i) & (ii) we get

$$f(x) + f(x+3) = 0$$

$$\Rightarrow f(x) = -f(x+3)$$

$$x \rightarrow x+3$$

$$\Rightarrow f(x+3) = -f(x+6)$$

$$\Rightarrow -f(x) = -f(x+6)$$

$$\Rightarrow f(x) = f(x+6)$$

$\Rightarrow$ Period of $f(x)$ is 6

$$\text{Now } f(-5) = f(5) = f(5-18) = f(-13) = 4$$

$$\& \quad f(1) = f(-1) = f(-1-12) = f(-13) = 4$$

$$\Rightarrow f(-5) + f(1) = 8$$

2. Ans (1)

$$f(x) = \frac{x+a}{x+b}$$

$$f^{-1}(x) = \frac{bx-a}{1-x} = \frac{a-bx}{x-1}$$

$$f(x) = f^{-1}(x) \Rightarrow b = -1$$

3. Ans (2)

$$x > 0$$

$$\tan^{-1}x + \cot^{-1}(-x) = 2\tan^{-1}6x$$

$$\Rightarrow \tan^{-1}x + \pi - \cot^{-1}x = 2\tan^{-1}6x$$

$$\Rightarrow \frac{\pi}{2} + 2\tan^{-1}x = 2\tan^{-1}6x$$

$$\Rightarrow \tan^{-1}6x - \tan^{-1}x = \frac{\pi}{4}$$

$$\frac{5x}{1+6x^2} = 1 \Rightarrow 6x^2 - 5x + 1 = 0$$

$$\Rightarrow \alpha = \frac{1}{2}, \beta = \frac{1}{3}$$

$$x < 0$$

Not possible

4. Ans (2)

$$\text{Let } g(x) = 3x^2 + (k-1)x + 3; x \in \mathbb{R}$$

$$\therefore \text{Range of } g(x) \in [0, \infty)$$

$$\text{so, } D = 0 \Rightarrow (k-1)^2 - 4 \cdot 3 \cdot 3 = 0$$

$$k-1 = \pm 6$$

$$\therefore k = 7, -5$$

5. Ans (7)

$$T_1 = 7120$$

$$T_2 = 7^3 + 1^3 + 2^3 = 352$$

$$T_3 = 3^3 + 5^3 + 2^3 = 160$$

$$T_4 = 1^3 + 6^3 = 217$$

$$T_5 = 352$$

$$\therefore T_{3n} = 160, n \in \mathbb{N}$$

$$\therefore T_{102} = 160.$$

6. Ans (1)

$$a_{n+1} = \left(a_n - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

So, $a_n \neq 0$ for $n \in \mathbb{N}$.

Also, a_n is an increasing sequence.

$$a_{n+1} - 1 = a_n^2 - a_n$$

$$\frac{1}{a_n - 1} - \frac{1}{a_n} = \frac{1}{a_{n+1} - 1}$$

$$\sum_{n=1}^{\infty} \frac{1}{a_n} = \sum_{n=1}^{\infty} \frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1}$$

$$= \frac{1}{a_1 - 1} - \left(\lim_{n \rightarrow \infty} \frac{1}{a_{n+1} - 1}\right) = 1$$

$$\therefore \lim_{n \rightarrow \infty} a_{n+1} \rightarrow \infty$$